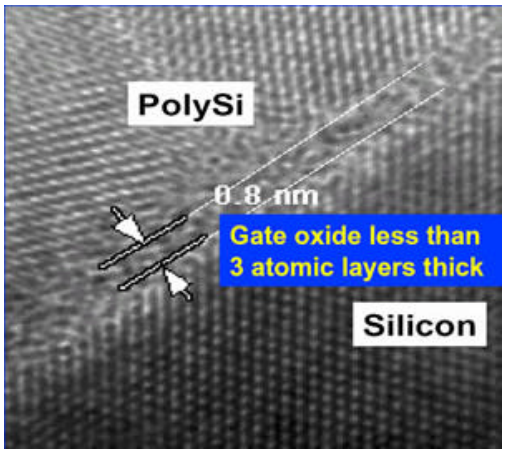
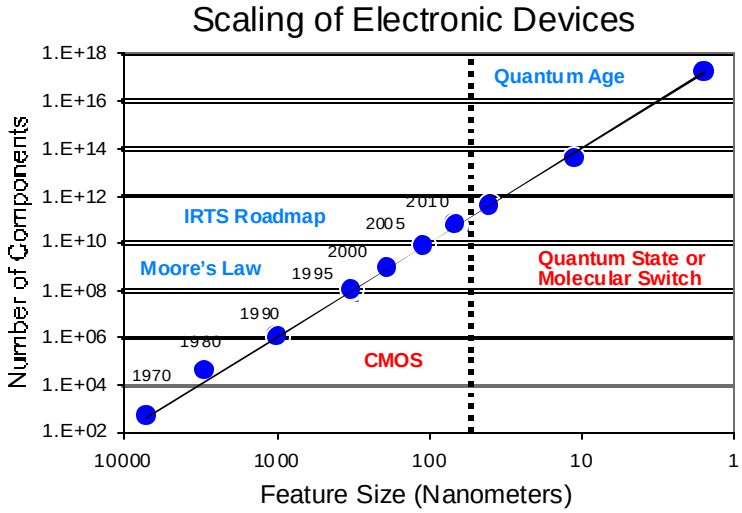
The Integrated Circuit Industry in 2003

Richard P. Rumelt



When Moore's Law Hits the Wall?



Moore's illustration of 3 atom gate oxide layer; Chart adapted from Dr. Karen H. Brown, Acting Director, National Institute of Standards and Technology, February 26, 2001

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By 2003, nearly 50 years had passed since the first transistor, 40 years since the planar process was developed for the production of integrated circuits (ICs). The IC industry had changed the world, yet its internal dynamics continued to flow from basic scaling rules and the demand generated by ever more powerful and ever less expensive chips. But in 2003, a number of strong forces were at work that seemed to mark the kind of "inflection point" that Intel's former Chairman, Andy Grove, had argued shaped the destiny of industries and firms. In particular, these four changes were challenging the idea of "business as usual:"

1. Total revenues in the semiconductor industry had grown at an average compound rate of 17% during the period 1970 to 2002. However, net growth over the most recent 8 years was nil. Some analysts believed that the slowdown was permanent and that the industry was finally reaching maturity.
2. Moore's law was based on continually reducing the size of the elements of the transistors in an integrated circuit. However, the physical world imposed certain limits. In particular, elements could not be smaller than atoms and signals could not travel faster than light.
3. Whereas finer and finer photolithography increased the density of transistors per square millimeter of silicon, each new generation of technology had dramatically increased the cost of the capital equipment needed for its implementation. Even if technological progress continued unabated, the cost of that progress might become prohibitive.
4. The rise of semiconductor foundries and fabless "manufacturers" might be signaling a dramatic shift in the structure of the industry.

Slowing Demand Growth

The worldwide IC industry had sales of about $150B in 2001, about 0.5% of world GDP. (This was about the same as GM's revenue for the year, and quite a bit less than Wal-Mart's.) Though sizable, the industry was most remarkable for its sustained growth and for its dynamism. Industry revenues had grown at an average annual rate of 17% over the past 25 years. Exhibits 1-3 provide data on global industry sales and the breakdown of sales by type of device and by end market.

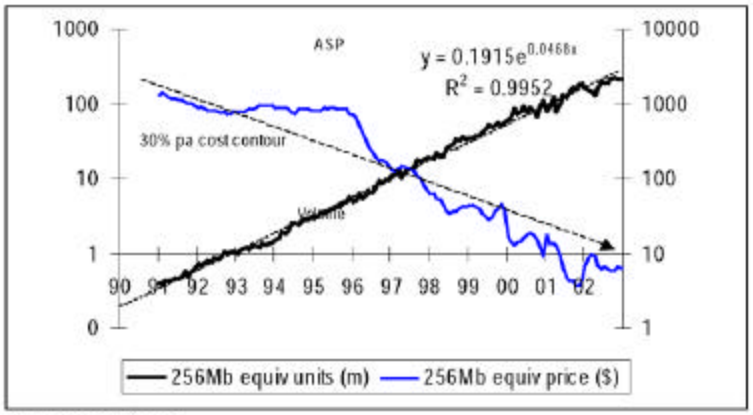
During the same 25 year period, the cost and price of a single transistor element of an IC had dropped at an average rate of 27% per year. The rise in revenues in the face of constantly deflating prices meant that unit volume (transistors/year)[2] had grown at an average annual rate of over 45%. However, in recent years (Figure 1), prices had fallen more rapidly than in the past yet unit demand had grown more slowly.

The semiconductor industry was global in extent, with leadership moving back and forth between the U.S. and Japan several times in its history. In 2003 it was about equally split among four regions: the Americas, Europe, Japan and Asia-Pacific. The most no-

[2] Counting chips as the unit of output distorted the picture because the functionality of a chip was determined by the number of transistors it contained. Transistors per chip had been growing at a compound rate of about 40% per year.

ticeable trend since 1991 was Japan had lost almost 50% of its semiconductor share to other Asia-Pacific competitors.

Figure 1
DRAM Price and Volume Trends
256Mb of Capacity



Source: WSTS, ML Research

The period 1996-2002 had seen a dramatic slowdown in the industry's revenue growth [Exhibit 1]. Industry revenues were $144B in 1995, $149B in 1999, spiked up to $204B in 2000, and then fell to and $139B in 2002. With five out of seven years having growth rates under 5%, many analysts believed that the days of rapid revenue expansion were finished. A minority predicted a rebound, arguing that flat revenues in the face of unit growth were due to extreme price-cutting, an effect of overcapacity. For example, from 1982 through 1995, DRAM average selling price per fbit (ASP/bit) declined 29% per year. By contrast, from 1995-2001, the rate of price decline was 45% per year. AMD Chairman W. J. ("Jerry") Sanders III described the situation as follows:

> It is now clear that 2000 was a bubble that distorted the industry's compound annual growth rate, pushing it above 16 percent... We believe that compound annual growth rates in the range of 8 to 10 percent will be the norm going forward over the longer term. This will represent a sea change for our industry. We can no longer count on the proverbial rising tide that lifts all boats. Nevertheless, a huge industry with compound annual growth averaging 8 to 10 percent will continue to offer opportunities for significant growth for companies that consistently deliver desirable products for their target customers.

IC Demand

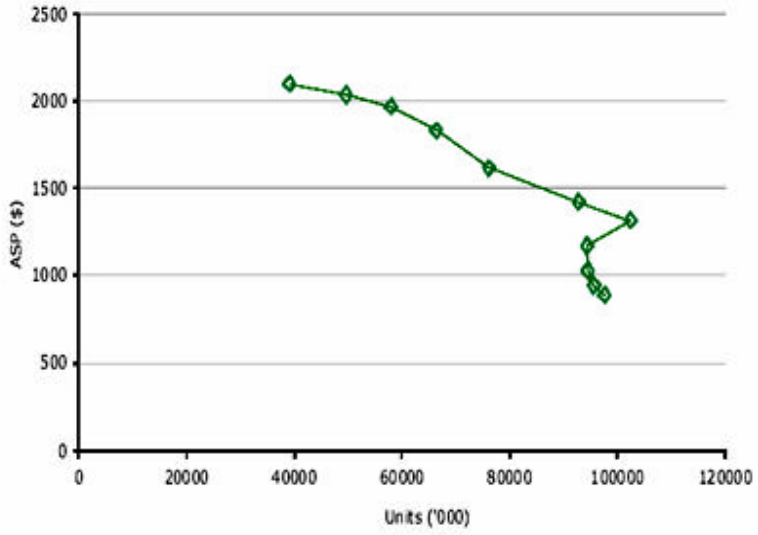
IC demand was indirect demand, being driven by the primary demand for the end products created with ICs. Most modern electronic products were built out of a collection of ICs. Exhibit 4 shows a teardown costing of an Ericsson mobile phone, identifying each IC in the device. A critical feature of modern electronic products was that an increasing fraction of their value was in ICs. As ICs became more capable, more and more of the value of the product lay in the logic and functions embedded in silicon. A simple overall view of IC usage by market is given in Exhibit 3. Data on breakdown by device-type is provided in Exhibit 2.

The most significant factor in the slower growth in IC demand was the maturing of the PC business. The PC-boom had been powered by the IC. ICs placed in PC and other computers accounted for 47% of IC revenues in 2001, down from 55% in 1998. Revenue growth in PCs and servers during 1997-2002 had been essentially zero, down from 16% per annum during the previous five years. The slower growth of the PC business was generally attributed to simple maturation: everyone who wanted a PC had one, and the pressure to upgrade had lessened. In addition, no new "killer" applications had appeared. Such applications, like desktop publishing, or the Internet, had generally cre-

ated the need for more complex software and hardware, spurring new purchases in the PC industry.

The flat revenue growth in PCs was interpreted by some as a fall in price elasticity. During 1992-97, ASP fell at an average rate of 4.7% while unit growth was 21.8% per year. However, during the next five years, ASP fell even faster, at 9.5% per year, but unit growth slowed to 10%. Most analysts forecast that PC would continue to account for a major, but slowly declining, portion of the IC market.

Figure 2
Price and Volume for Desktop PCs



Source: ABN AMRO

The data networking equipment sector was also an important user of ICs, and had grown rapidly during the 1990s. The Y2K upgrade push, the Internet, and the migration to LAN networks all contributed to expansion. However, with LAN penetration at 70% and the bursting of the Internet bubble, further growth in this sector would likely be based upon normal replacement cycles rather than new applications.

In telecommunications equipment, the 1999-2000 frenzy of expansion had fallen off sharply in 2001-2002. Equipment sales of $14B in 1998 had jumped to $27B in 1999 and $45B in 2000, only to fall to $32B in 2001 and $18B in 2002. Optimists saw a recovery in this sector in 2003, whereas pessimists believed that incumbents had little incentive to re-equip and next-generation players were largely out of business.

Opinion on the future was divided. SG Cowen, for example, recommended that the information technology industry get used to "life in the slow lane." One reason for this position was that the IT industry had become, over the past 40 years, very large. In 2001 it accounted for one-quarter of all private capital investment, up from 6% in 1960. In addition, SG Cowen noted that growth was limited by new applications and new users:[3]

> First, there simply aren't that many new users to find. Fueled by the Internet, the penetration of computers in businesses and homes has risen dramatically

[3] "Tech Roadmap 2003," January 2003, p. 19.

> over the past several years. Markets in the leading industrial countries are largely saturated, with computers in widespread use within virtually all businesses and a near majority of households. For example, in the U.S., PCs are now in use in 60% of all homes and over 80% of those households with annual income of $50,000 or more. In less developed countries, penetration is decidedly lower, particularly in homes, but lower standards of living are likely to limit the speed and depth of adoption….
>
> Second, we've already automated the most critical tasks. Think about it. In the past five years, you and almost everyone you know started using email, surfing the Web and shopping on-line. The change in your use of computing between then and now is significant. But now that you're on-line, how much more are you likely to do? You're no longer a new user—the only growth you generate comes from spending more time or doing more on line. And, unless you're hooked on downloading music or playing games, the reasons to spend that time are limited. Businesses are in the same boat. Most of them spent the last several years getting their employees and partners connected and tackling the easiest applications. They will undoubtedly add more applications from here, but these investments won't drive the same sort of growth as the original build out of infrastructure and applications.
>
> Finally, the next big application isn't very obvious. We've spent the last couple of years looking at a number of candidates, and most of them are either too nascent or too limited in scope to make a big difference. A good example is web services. Historically, every big application cycle has been enabled by a new approach to application development. By enabling the integration of widely distributed information and processes through industry standard, web services offers just such a new approach to development. Unfortunately, most of the focus of web services has been on integrating our existing systems. We have yet to hear about the great new application that we will be able to create with web services that we haven't seen before, and without that, it's hard to imagine web services driving a dramatic expansion in the market….we still haven't seen anything that has triggered the same "ah ha" we got when we first used a spreadsheet, surfed the web, or shopped at Amazon.com.

Other analysts remained optimistic about the future of the IC industry. For example, Merrill Lynch argued that the IC industry wasn't really tied to information technology:[4]

> Readers who think our end market analysis is too optimistic should take note of inherent risk of understating semiconductor growth forecasts based on end market growth. As we've said, the unique thing about the semiconductor industry that it creates new markets through cost reduction, performance improvement, and power-consumption reduction.
>
> Historically, trying to predict semiconductor growth from end market growth has always understated future prospects because semiconductors grow faster than their underlying end markets. No one in the early 1980s would have predicted the growth of the PC (Bill Gates famously said in 1981 that he never expected PC memory to exceed 640kB – today's consumer PCs have 4,000 times more memory), nor in the early 1990s the digital wireless boom, nor the digital consumer explosion of set-top boxes, DVDs, computer games, and digital cameras. Semiconductors do actually grow faster than the end markets they serve.

[4] "Return of the Chip," September 2002, p. 142.

An additional source of uncertainty was demand in the developing world, especially China. IC Insights forecast that China would account for 20% of the total IC market by 2007, up from 12% in 2002.

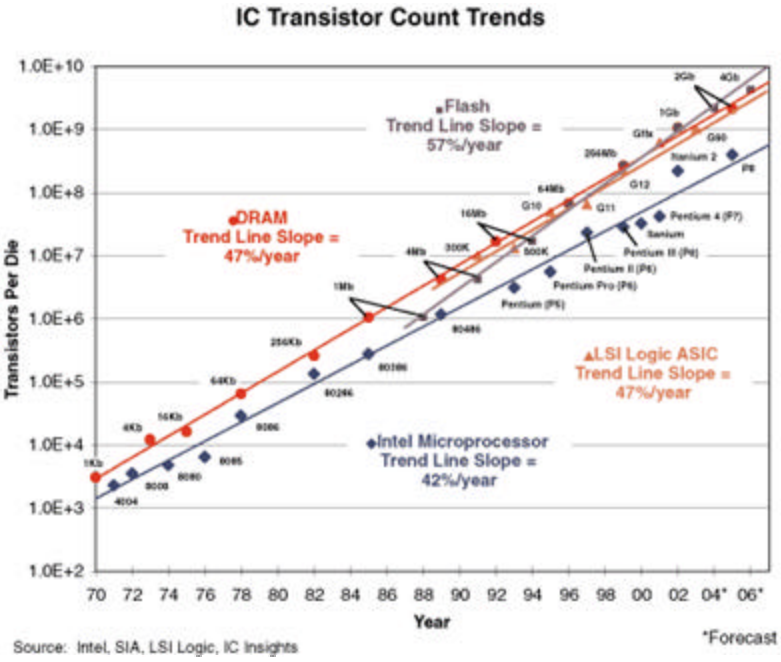
Limits to Scaling Down?

In 1965, Gordon Moore, one of the developers of the integrated circuit, noted that the "complexity for minimum component costs has increased at a rate of roughly a factor of two per year". That is, each year, efficiently produced chips doubled in their capabilities. This continuing progress, known as "Moore's Law," powered the development of the integrated circuits industry and transformed the world. Personal computers, mobile telephones, smart appliances, smart bombs, GPS navigation, digital television, the Internet, and a host of other new industries and technologies had been created by the continuing progress in integrated circuits.

Twenty two years later, in 1997, Gordon Moore, speaking to the Intel Developer Forum, said that the IC industry's ability to keep doubling the power of a chip every 18-24 months was running into physical limits. The chairman emeritus of Intel showed the Forum an image of a microprocessor made with 250 nm technology in which it was possible to see and count the individual layers of atoms. The essential granularity of the physical world implied, he noted, that it would be impossible to keep moving ever smaller feature sizes. "Some time in the next several years we get to some finite limits, but not before we get through five generations," Moore observed. Looking ahead, five generations of technology would predict that the industry would hit the line-resolution "wall" in 2017.

The progress made by the semiconductor industry required technologies that coordinated with each other. Each decrease in feature size required new lithography equipment, mask creation systems, design-support software, manufacturing processes, interconnect technologies, etc. No one firm possessed cutting edge know-how in all these technologies, with the line between the fabricators and the capital goods manufacturers being especially clear. To deal with this coordination problem, the industry had developed a "roadmap" describing the expected path of technology evolution up to 15 years into the future.

Figure 3



The first roadmap had been created by the Semiconductor Industry Association (SIA). Its efforts were expanded to the global industry in 1998, leading to the creation of the International Technology Roadmap for Semiconductors (IRTS). The 2001 IRTS predicted industry needs

and potential solutions through 2016. The IRTS Roadmap was not a projection based on known technology. Rather, it was a "challenge" for the industry to keep fulfilling the promise of Moore's Law. The report's introduction began with a warning:

> The ITRS time horizon (15 years) provides a limit to what may be considered "on/off the Roadmap." To date, each edition of the ITRS has been built around a view toward continued scaling of CMOS (Complementary Metal-Oxide-Silicon) technology. However, with the 2001 edition, we are reaching the point where the horizon of the Roadmap challenges the most optimistic projections for continued scaling of CMOS (for example, MOSFET channel lengths of roughly 9 nm). It is also difficult for most people in the semiconductor industry to imagine how we could continue to afford the historic trends of increase in process equipment and factory costs for another 15 years!

The IRTS 2002[5] update did not accelerate any of the 2001 targets, a first in the construction of such roadmaps. Exhibit 5 displays key elements of the 2002 IRTS roadmap, together with other data.

The key driver in the roadmap was the concept of a "technology node." The node, in turn, was defined in terms of the "feature size"? half the distance beween cells in a dynamic RAM memory chip. Transistor density was proportional to the inverse of the square of feature size. Thus, a reduction of feature size by a factor of 0.7 implied a reduction in transistor area by a factor of 0.49, a doubling in transistor density. Historically, during 1980-1995, new *technology nodes* representing a scaling factor of 0.7 had occurred every 3 years. Historically, a *generation* had been considered to be *two nodes*. Thus, feature size was halved each generation, producing a quadrupling in density (thus, generations of DRAM had bit capacities of 1M, 4M, 16M, and so on). During the period 1995-2001, technology had moved faster, with new 0.7 nodes introduced every 2 years. The IRTS Roadmap projected a return to 0.7 3-year pattern, implying a halving of feature size (4X density) every six years (a decline of 11% per year). Thus, the Roadmap showed the feature size for DRAMs falling from 130 nm in 2001 to 65 nm in 2007.

The history of the IC industry was replete with warnings that the limits to progress were soon to be reached—usually in "about 10 years." In 1987, an article in Scientific American noted that Moore's Law would probably not continue through the end of the 1990s. Nevertheless, a single silicon atom was approximately 0.2 nm in size, and as feature sizes fell below 90 nm, key elements (i.e., gate width) would be only a few nanometers wide. At these extremely small sizes, statistical and quantum effects began to bring wholly new issues into play. Some industry scientists were optimistic, but others believed that the optimism reflected past successes and the considerable investment in existing knowledge. James Heath (UCLA), co-inventor of the Buckyball molecule, said "I think their optimism for being able to continue until 2014 is not very realistic. When you get to very, very small sizes, you are limited by relying on only a handful of electrons to describe the difference between on and off."

Despite Moore's warning, or perhaps because of it, Intel was more committed than ever to pushing the boundaries of silicon technology. In early 2001, Intel researchers built a 20-nanometer transistor, announcing that it "demonstrated that there are no fundamental barriers to extending Moore's Law for another decade." At production scale, such transistors would allow 20 Ghz microprocessors, containing one billion transistors, and operating at less than 1 volt. Dr. Gerald Marcyk, director of Intel's Components Research

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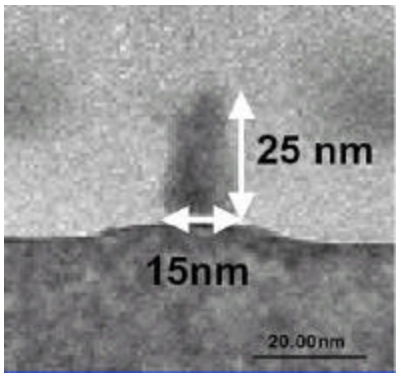
[5] http://public.irts.net.

Lab, believed that Moore's Law would hold for another three technology nodes or more. In early 2002, Intel researchers announced an experimental 15-nanometer transistor (Figure 4).

Figure 4
Experimental 15 nm Gate-width MOS Transistor



Source: Intel

To push Moore's Law, there were a number of technological avenues that the IC industry could follow. To continue to work with silicon, many technologists believed that "photo-lithography" would have to move to ultra-violet and x-ray methods, considerable steps into the unknown. After that, chips would have to be built in more complex layers, with more complex interconnections. Such an approach would overcome the density problem, but would also undermine the basic driver of Moore's Law: the simplicity of planar technology, wherein transistors were constructed by photolithography rather than as complex 3-dimensional devices. Furthermore, at very small feature sizes, extremely tiny particles, which had not been a source of defects in the past, would have to be eliminated to keep yield rates at economic levels.

Related to the issue of scaling down was that of power density. Increased transistor density meant increased power density. Whereas an Intel 386 processor (1988) generated about 2 watts/cm^2, a 2001 Pentium 4 generated about 40 watts/ cm^2. Exponential growth in this parameter could not be supported: the silicon would simply melt. The power density of a nuclear reactor was about 500 watts/cm^2, and that of a rocket nozzle was about 2000 watts/ cm^2. The traditional solutions to the heat problem were to use more active cooling and to lower the operating voltage. Beyond a certain heat density, expensive and bulky refrigeration machinery would be required for active cooling. Reducing voltages would continue to work for a while, but as operating voltages dipped below 0.5 volts, random effects would begin to overwhelm the difference between "ON" and "OFF" states.

Beyond the realm of silicon lay the worlds of quantum computing, DNA-computing, and nanotubes. Some early results in these fields were promising, but nothing approaching a useful product had yet been demonstrated. By analogy to the semiconductor world, these technologies were in the pre-transistor phase of development.

Rising Capital Costs

Moore's "Second Law" was that the cost of developing and building a fab doubled with each new generation. Most industry analysts had generalized this "law" to the simpler idea that fab costs increased exponentially over time. The contentious issue was whether the rate of cost increase would be faster or slower than the gains in chip performance. For example, doubling the number of transistors on a wafer required more expensive capital equipment. If the rise in equipment cost was more than 2X, the economics would begin to work against the technical progress. Moore himself took the position that things were beginning to run the wrong way:[6]

> In today's dollars, the cost of a new "fab" has risen from $14M in 1966 to $1.5B in 1995. By 1998 work will begin on the first $3B fabrication plant. Between 1984 and 1990, the cost of a fab doubled, but chip makers were able to

[6] Economist 1995.

> triple the performance of a chip. In contrast, the next generation of fabs will see cost double again by 1998, but this is likely to produce only a 50% improvement in performance. The economic law of diminishing marginal returns appears to be setting in.

The capital cost of building and equipping a semiconductor fabrication facility ("fab") had grown exponentially over time. In 1970, a fab cost about $10-20 million to build; the 300 mm wafer fabs coming on line in 2002-2002 cost an average of $2.6 billion, and there were $4 billion fabs in the planning stages (300mm wafer, 90 nm technology).

Goldman Sachs analysts noted that whereas industry revenues had increased fivefold in the past 10 years, the cost of building a fab had increased sevenfold. Fabs were becoming more expensive because equipment was more expensive and because equipment was an increasing portion of the expense. In the 1980s, production equipment accounted for about one-half of total capital expenditure on a fab. By 2001, it had risen to 70% of the total. The equipment itself had to process higher resolutions, clean rooms had to meet ever higher standards, there were an increasing number of masking steps and manufacturing process steps with each generation, and the handling and testing processes were more and more complex.

The rule of thumb in the industry was that a fab had to generate annual revenues at least equal to its capital cost to generate a reasonable return on investment. With new cutting edge fabs costing $4 billion, few firms in the industry had revenue streams sufficient to justify such an expense. The largest foundry in the world, TMSC, had revenues of about $5 billion in 2002.

There was no simple summary of the history of capital costs in the industry. Each fab built was different, and statistics gathered and reported conflated fabs on the drawing board with fabs operating at peak capacity, designed years before. Furthermore, fabs built for DRAM production were generally less expensive than those built for the manufacture of microprocessors. Location also played a significant role in fab cost. Exhibit 6 provides a summary of historical DRAM fab capital costs and also shows the trend in line-width (feature size) over time.

As Exhibit 6 shows, the capital costs of a fab had risen roughly at the rate of 16% per year. Most industry analysts predicted future growth in costs of about 9% per year, in line with a somewhat slower rate of progress in transistor density and wafer size. Other analysts, however, continued to use the 16% rule, drawing the conclusion that fab costs would shut the industry down by 2017. A third approach to the forecasting problem was to relate fab costs to technology rather than time. Exhibits 7a and 7b displays fits to exponential time trends DRAM fab costs and for feature size.

Rising fab costs made for a brisk demand for capital equipment. Exhibit 8 displays the histories of the IC industry's capital spending and capex/revenues ratio. Capital spending by IC manufacturers quite cyclical, responding to the primary demand growth as much as to technology advances. Rising fab costs were also simply a reflection of rising fab capabilities. In 2002, a new fab's capital costs equaled about 1.3% of total industry revenues. If fab costs escalated faster than revenues, that ratio would increase. Each new fab would have to take an increasing share of the total industry's activity to be economic and each new fab would be an increasingly significant addition to capacity.

Do "Real Men" Need Fabs?

In the late 1980s, W.J. Sanders III, chairman of Advanced Micro Devices (AMD) coined the phrase "real men have fabs." The implication was that serious IDMs did their own

manufacturing, a viewpoint going back to the beginnings of the semiconductor industry. That viewpoint was based on the proposition that there was a close connection between the knowledge used in design and manufacturing. To push the frontier, one had to know the feel and contour of the "edge."

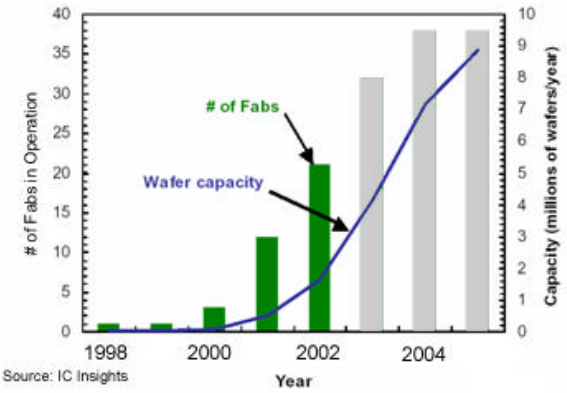
Nevertheless, early in 2002 AMD announced a joint venture with United Microelectronics Corp. (UMC, Tawian) to own and operate a state-of-the-art 300-mm wafer fab in Singapore to make microprocessors and other products. Explaining his change of heart (or of gender), Sanders said "Today's agreement reflects an innovative response to the tectonic shift that has changed the fundamental economics of the worldwide semiconductor industry." Robert Tsao, chairman and chief executive officer of UMC, said "This is the first agreement under which a premier foundry company joins forces in a broadly unified partnership with a leading integrated device manufacturer. I expect our collaboration will set the example for success between pure-play foundries and leading semiconductor manufacturers."

The first pure foundry was Taiwan Semiconductor Manufacturing Co. Ltd. (TSMC). In subsequent years, UMC and Chartered Semiconductor (Singapore) were formed. IBM also acted as a foundry, specializing in high-performance silicon germanium products. Foundries manufactured ICs designed by others. An IC company that did its own design and manufacturing was called an "integrated device manufacturer" (IDM). One that used the services of a foundry was called 'fabless." Some IDMs were fabless for certain products, and increasing numbers were going the fabless route entirely.

Foundries blossomed during the 1990s as the costs of new fabs skyrocketed and brisk demand pushed many firms to seek ways to increase capacity rapidly. In addition, the general dominance of CMOS technology over other forms of semiconductor manufacturing meant that IDMs placed less emphasis on process differentiation. Several examples also spurred interest in the fabless-foundry business model. Qualcomm introduced its CDMA wireless technology in 1995, outsourced manufacturing, and became the largest fabless company in the world, with $1B in sales of CDMA chips in 1998. Nvidia used a fabless strategy to win out over at least 10 other firms in the video board market, taking the leading position.

The leading foundry was TSMC with $5 billion in revenue in 2002, up 36% over 2001, and constituting 54% of the world-wide foundry market. The runners up were UMC and Chartered.

Figure 5



The most carefully watched new entrants in the foundry business were located in China: Semiconductor Manufacturing International, Grace Semiconductor Manufacturing, and Advanced Semiconductor Manufacturing, all in Shanghai. These firms had very good capacity utilization rates and showed good growth. On the other hand, industry experts noted that they were far from the leading edge. As one put it, "Fabless guys want leading edge technology." But firms like CSMC were using old technology and 5 inch wafers. They were presently focused on sup-

plying low-tech ICs for simple consumer products, but had little appeal to Western fabless competitors.

Fabless firms faced additional choices about how much integration to undertake. Some fabless firms produced their own tooling (for masks and packaging), whereas others contracted that out as well. In addition, a company could employ an electronic design services firms (EDS) to do actual IC design, and electronic design automation (EDA) firms, to work out prototyping, tooling, and manufacturing issues. Finally, as the software used to design an IC became more and more capable, design of non-cutting edge devices had become more and more routine.

The key advantages claimed for a foundry were that it did not have to invest in creating designs, but could concentrate on manufacturing efficiency. Furthermore, a foundry spread cyclical risk over a variety of product lines, maintaining a higher rate of capacity utilization than the average IDM. With regards to the advantages for the fabless IC companies, IC Knowledge stated:

> - The start-up of a fabless semiconductor company requires only a relatively modest investment in design systems and avoids the enormous cost of building a semiconductor manufacturing facility, a cost that now exceeds $2 billion for state-of-the-art "fabs".
> - The fabless semiconductor companies reap the benefits of manufacturing economies of scale by manufacturing their products on lines running products from other companies.
> - The fabless company can focus all of its efforts in the area of its greatest expertise, product definition, design and development and not worry about manufacturing.
> - Fabless companies can also pick from the whole range of processes available from foundries as a group, as opposed to being tied to what is available for processes at an internal fab. Of course IDMs can go outside for processes and increasingly are doing so, but when your in-house fabs are underutilized there is a great deal of pressure to fill them even if the fit isn't great.

Many industry observers believed that in the DRAM business, only a few firms, principally Samsung and TI, would continue to build their own fabs, with middle-sized players forced out of business or into foundries.

The future development of the fabless-foundry business model was the most controversial subject in the industry. The foundry industry had grown at a rate of 31% per annum over 10 years and many thought that it would account for 40-50% of the world IC industry's output (by value) by 2010. Some firms, like Nvidia and AMD, were betting on that future, sidestepping the problem of investing billions in new fabs. Other's warned against this development. One manufacturer noted that

> foundries have at times had more difficulty than autonomous manufacturers in adopting new technologies or manufacturing methods. TSMC's move from the 180-nanometer chip-making process to the 130-nanometer process proved difficult and contributed to delays for products from Transmeta and others.

It was also argued that designers would lose their technical edge. CNet Asia quoted Syed Ali, CEO of Cavium Networks, as saying "I have not yet seen anyone who has taken a 0.13-micron process (from a foundry) and done well. Many [fabless] chip start-ups are staffed with engineers who have never seen a transistor under a microscope."

First Albany, an institutional equity research firm, argued that ventures between IDMs and foundries to develop new cutting edge fabs would fail:

> Most, if not all, of the joint process developments between device makers will fail as they have in the past. We cannot point to one manufacturing partnership that has succeeded in the semiconductor market…. The foundries can no longer be fast followers as they were in the 1990s. The pure-play foundries do not have the cash flows to support the required R&D and capital spending to compete in the leading-edge logic market. Increasingly, we see foundry services becoming a commodity much like DRAMs. We believe that this will doom the device makers that depend on them to lose share in leading-edge markets.

First Albany's analysis drew from their belief that the industry transition from 180 and 130 nm technology to 90 nm would be critical. Intel was well in the lead in this transition, and First Albany argued that this was a direct result of their continued deep involvement in production. Without production, they argued, a firm might have a brief win in the marketplace, but would be unable to build a sustainable advantage. By contrast, they argued, "the first companies to 90nm and a working low-k dielectric process will have a significant competitive advantage that will drive market share gains."

The changing economics of the IC industry led some observers to conclude that consolidation was a likely result. The combined effect of slower growth and higher investment costs led many analysts to conclude that only a few large players would prosper. SG Cowen's view was typical: "Faced with these modest growth prospects, the industry will likely undergo significant changes over the next few years, including consolidation around the leading vendors, commoditization of many core technologies, and a blurring of the traditional lines between vendors and categories."

Consolidation had already taken place in the DRAM industry. The entry of Samsung, and other Korean manufacturers, in the 1990s, created overcapacity, pricing pressure, and consolidation, with Samsung emerging as the share leader (26%). The consolidation wave also saw Idaho-based Micron Technology acquire TI and Motorola's DRAM capacity and then buy Toshiba's DRAM operations in late 2001. NEC's and Hatachi's DRAM operations were combined to form Elpida Memory, and Hynix was formed by the combination of Hyundai and LG Semicon.

First Albany[7] saw slower growth and increased capital requirements as pressing all but the strongest IC companies to the edge.

> The industry is facing a very expensive and very complex transition. After five years of sub-par growth and lackluster profits, companies are now faced with the transition from 200mm to 300mm wafers…. We believe that few companies will be able to stay on the semiconductor technology treadmill. Many are deferring investments and increasing the amount of outsourcing to satisfy incremental demand at the leading edge.

The one firm that First Albany saw staying ahead of the curve was Intel.

A different stance was taken by Merrill Lynch, which believed that the industry was deconsolidating.[8] They stated that "the growth deceleration of the Japanese semiconductor leaders of a decade ago has left a void. The top 10 players now control less of the mar-

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[7] "Semiconductor Consolidation," April 2002.

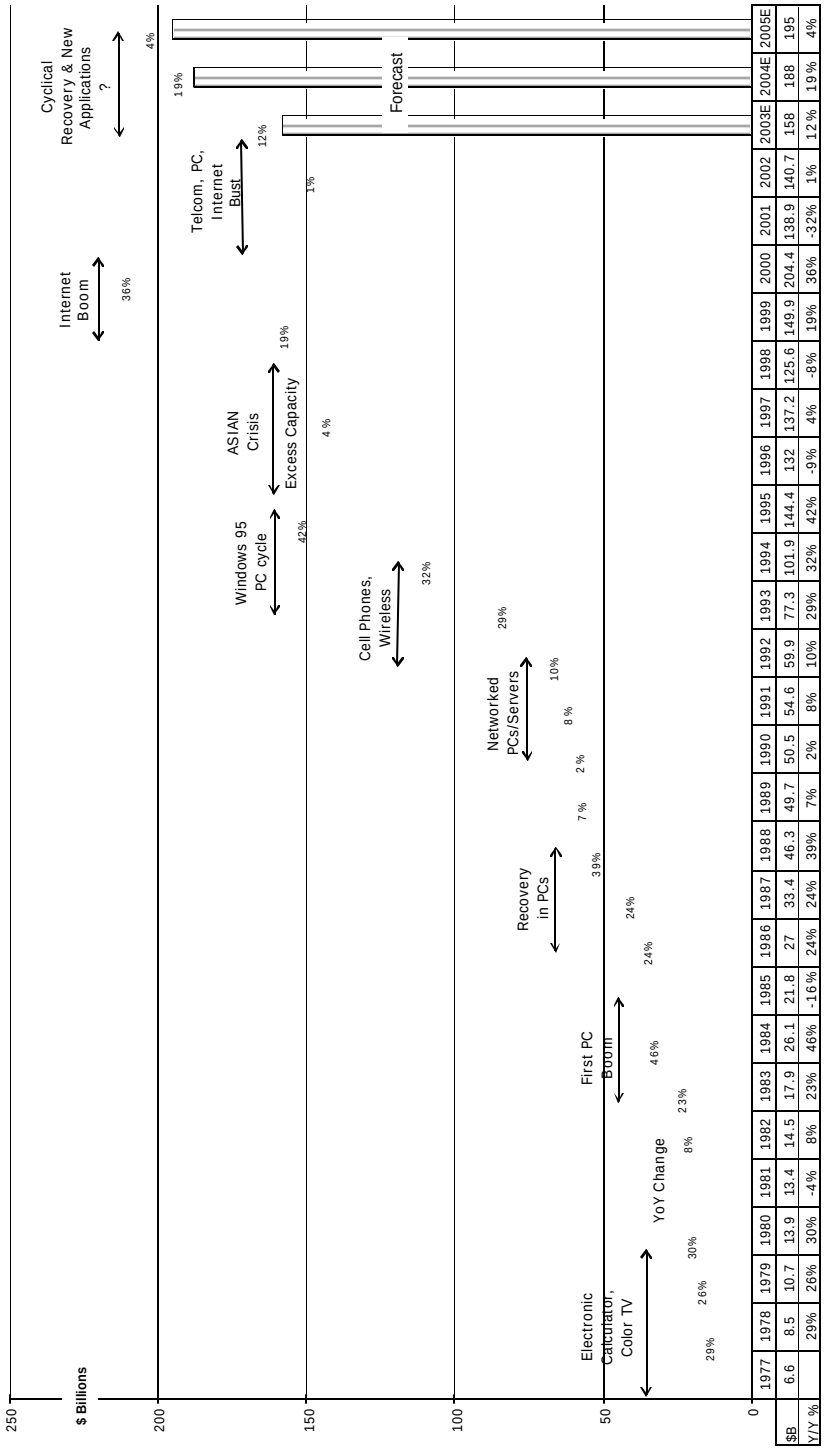
[8] "Global Semiconductor Investing: Return of the Chip," 6 September, 2002, p.4 & 157.

ket than a decade ago." Indeed, the share of the top ten had fallen from 63% to 48% in a decade. Continuing the argument, Merrill Lynch said

> This is highly counter-intuitive, because the industry keeps telling us that higher production plant costs (called fabs in the trade) will lead to consolidation. Foundries are not the reason. Even though foundries have made the small semiconductor company business model a reality, the capital spending share of the top ten capital spenders is also shrinking. We were very surprised by this finding and expect at least the capital spending concentration trend to reverse in the coming decade, driven especially by a move to 300mm capacity that few companies can afford….The deconsolidation of semiconductor suppliers is only part of a much broader disaggregation of not just the semiconductor industry, but also the entire electronics industry. Chart 358 describes how integrated semiconductor companies first outsourced their manufacturing equipment needs, then design software, manufacturing itself, and finally semiconductor intellectual property (IP).

Exhibit 9 displays Merrill Lynch's view of the general disaggregation occurring in the IC industry.

Exhibit 1: Global Indusgtry Revenue



	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003E	2004E	2005E
$B	6.6	8.5	10.7	13.9	13.4	14.5	17.9	26.1	21.8	27	33.4	46.3	49.7	50.5	54.6	59.9	77.3	101.9	144.4	132	137.2	125.6	149.9	204.4	138.9	140.7	158	188	195
Y/Y %		29%	26%	30%	-4%	8%	23%	46%	-16%	24%	24%	39%	7%	2%	8%	10%	29%	32%	42%	-9%	4%	-8%	19%	36%	-32%	1%	12%	19%	4%

Source: Adapted from JMP Securities, “Semiconductors: The Long and Winding Road to Recovery,” Nov. 5, 2002.

Exhibit 2: Semiconductor Sales by Type of Device
($ in millions, units in millions, price in $)

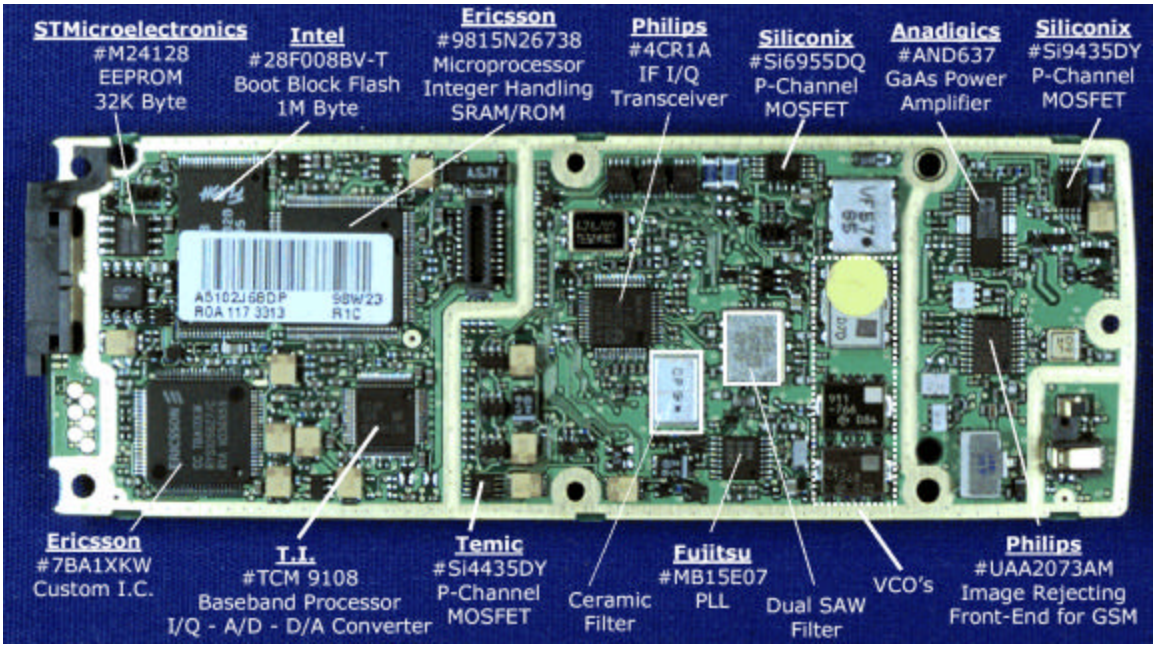
								CAGR
Year		1990	1993	1995	1999	2000	2001	1990-
Discrete	Rev	7,338	8,637	14,005	13,089	16,923	12,185	4.7%
	Vol	94,553	108,226	161,067	208,395	257,988	201,314	7.1%
	ASP	0.08	0.08	0.09	0.06	0.07	0.06	-2.2%
Opto	Rev	2,118	2,654	4,344	5,796	9,805	7,372	12.0%
	Vol	7,195	7,423	12,134	23,000	27,855	22,502	10.9%
	ASP	0.29	0.36	0.36	0.25	0.35	0.33	1.0%
Sensor	Rev	na	na	na	298	721	913	75.0%
	Vol	na	na	na	290	999	1472	125.3%
	ASP	na	na	na	1.03	0.72	0.62	-22.4%
Bipolar	Rev	4,059	3,150	2,774	991	1,069	439	-18.3%
	Vol	5,957	5,206	4,463	2,282	2,019	980	-15.1%
	ASP	0.68	0.61	0.62	0.43	0.53	0.45	-3.7%
Analog Std Linear	Rev	2,972	3,788	5,726	7,215	11,406	9,246	10.9%
	Vol	6,335	7,775	10,791	16,340	22,584	19,159	10.6%
	ASP	0.47	0.49	0.53	0.44	0.51	0.48	0.3%
Analog Application Specific	Rev	4829	6885	10920	14837	19110	13934	10.1%
	Vol	6608	7652	10316	15577	18679	14007	7.1%
	ASP	0.73	0.90	1.06	0.95	1.02	0.99	2.8%
MPU	Rev	2,234	8,590	14,279	27,099	31,881	23,354	23.8%
	Vol	113	167	212	326	385	293	9.0%
	ASP	19.77	51.44	67.35	83.13	82.81	79.71	13.5%
MCU	Rev	3,663	5,886	9,007	9,662	12,300	9,660	9.2%
	Vol	1,390	2,169	2,933	4,394	5,503	4,806	11.9%
	ASP	2.64	2.71	3.07	2.20	2.24	2.01	-2.4%
DSP	Rev	291	675	1,729	4,403	6,142	4,256	27.6%
	Vol	53	52	134	598	918	662	25.8%
	ASP	5.49	12.98	12.90	7.36	6.69	6.43	1.4%
Gen. Logic	Rev	4,756	6,168	10,743	12,043	15,058	9,136	6.1%
	Vol	na	na	na	12,734	16,833	11,530	na
	ASP	na	na	na	0.95	0.89	0.79	na
Field Prog Logic	Rev	916	1,257	1,647	2,999	5,456	2,097	7.8%
	Vol	na	na	na	444	522	233	na
	ASP	na	na	na	6.75	10.45	9.01	na
Spec Purp. Logic	Rev	5,562	8,354	15,773	18,635	25,298	21,494	13.1%
	Vol	na	na	na	6,199	8,239	7,165	na
	ASP	na	na	na	3.01	3.07	3	na
DRAM	Rev	6,356	13,140	40,833	20,743	28,907	11,199	5.3%
	Vol	1,315	1,499	2,477	3,239	3,995	4,060	10.8%
	ASP (1)	4.83	8.76	16.49	6.4	7.24	2.76	-5.0%
Flash	Rev	na	na	1,860	4,572	10,637	7,595	na
	Vol	na	na	235	1,225	1,866	1,437	na
	ASP (1)	na	na	7.93	3.73	5.7	5.28	na
Other Memory	Rev	5,423	8,127	10,765	7,006	9,683	6,082	1.0%
	Vol	1,796	2,559	3,381	4,801	4,972	4,215	8.1%
	ASP (1)	3.02	3.18	3.18	1.46	1.95	1.44	-6.5%
Total Semi	Rev	50,517	77,311	144,405	149,388	204,396	138,962	9.6%
	Vol	125,315	142,728	208,143	299,844	373,357	293,835	8.1%

Exhibit 3: Semiconductor Sales by End Market
($ in millions)

	1999	2000	2001	2002E
PC	31%	31%	30%	31%
Servers	3%	4%	4%	4%
Computer Peripherals	4%	3%	3%	3%
Cellphone/PDA/wireless basestn	15%	16%	16%	17%
Enterprise/SOHO networks	12%	13%	11%	9%
Public Wired Telecom	6%	8%	6%	4%
Analog Consumer Elect	6%	4%	4%	4%
Digital Consumer Elect/TV/LCD	2%	2%	4%	5%
Industrial/Instruments/Mil/Aero	16%	14%	16%	17%
Auto Electronics	5%	5%	6%	7%
Total	100%	100%	100%	100%

	1999	2000	2001	2002E
PC	$46.30	$63.40	$41.70	$43.20
Servers	$4.50	$8.20	$5.60	$5.80
Computer Peripherals	$6.00	$6.10	$4.20	$3.80
Cellphone/PDA/wireless basestn	$22.40	$32.70	$22.20	$23.30
Enterprise/SOHO networks	$17.90	$26.60	$15.30	$13.00
Public Wired Telecom	$9.00	$16.40	$8.30	$5.30
Analog Consumer Elect	$9.00	$8.20	$5.60	$5.30
Digital Consumer Elect/TV/LCD	$3.00	$4.10	$5.60	$6.70
Industrial/Instruments/Mil/Aero	$23.90	$28.60	$22.20	$23.20
Auto Electronics	$7.50	$10.20	$8.30	$9.60
Total	$149.40	$204.40	$139.00	$139.10

Exhibit 4: Teardown of Ecrcsson SH888 GSM Mobile Phone



Qty.	Mfr.	PN	Cost Estimates	
			Component	Insertion
1	Ericsson	9815N26738	$ 5.875	$ 0.200
1	Ericsson	7BA1XKW	$ 4.900	$ 0.200
1	Ericsson	TCM 9108	$ 2.650	$ 0.200
1	Philips	4CR1A	$ 1.763	$ 0.200
1	Intel	28F008BV-T	$ 10.349	$ 0.200
1	Anadigics	AND637	$ 4.180	$ 0.200
1	Philips	UAA2073AM	$ 0.767	$ 0.200
1	Fujitsu	MB15E07	$ 0.602	$ 0.200
4	Toshiba	7W08	$ 0.352	$ 0.200
1	Unknown	C70851	$ 0.586	$ 0.060
1	STM	24128W6	$ 0.878	$ 0.060
2	Siliconix	Si6955DQ	$ 0.489	$ 0.200
16	**TOTALS**		**$ 34.937**	**$ 2.920**

Source: Portelligent

Exhibit 5: IRTS 2002 Roadmap

History:			
Year	1995	1997	1999
DRAM Feature Size (nm)	**350**	**250**	**180**
Product Generation (Mbits)	4	16	64
Actual Mbits	4.2	16.8	67.1
Chip Size (Sq. mm)	35	45	88
Wafer Diameter (mm)	200	200	200
ASP cents/Mbit	328	44.7	16.4

Roadmap:										
Year	2001	2002	2003	2004	2005	2006	2007	2010	2013	2016
DRAM Feature Size (nm)	**130**	**115**	**100**	**90**	**80**	**70**	**65**	**45**	**32**	**22**
Product Generation (Mbits)	512	512	1000	1000	2000	2000	4000	8000	32000	64000
Actual Mbits	537	537	1074	1074	2148	2148	4296	8592	34368	68736
Cell Area Factor	8	8	6	6	6	6	6	6	4	4
Cell Area (Sq. Microns)	.13520	.10580	.06000	.04860	.03840	.02940	.02535	.01215	.00410	.00194
Area of All Cells on Chip (sq. mm)	72.6	56.8	64.4	52.2	82.5	63.2	108.9	104.4	140.8	133.1
Cell/Chip Area Ratio	57.2%	56.8%	54.6%	56.1%	56.1%	54.4%	59.5%	57.7%	58.9%	55.9%
Chip Size (Sq. mm)	127	100	118	93	147	116	183	181	239	238
Wafer Diameter (mm)	300	300	300	300	300	300	300	300	450	450
D0 for 90% Yield (faults/Sq. meter)	1963	2493	2148	2748	1752	2236	1426	1356	1356	1356
#Mask Levels	21	22	24	24	24	24	24	26	26	26
Voltage	1.1	1	1	1	0.9	0.9	0.7	0.6	0.5	0.4
ASP cents/Mbit	2.70	2.57								
MPU Gate Length (nm)	**150**	**130**	**107**	**90**	**80**	**70**	**65**	**45**	**32**	**22**
High-Volume MPU										
Trans/Sq. Cm. Logic (millions)	38.6	48.6	61.2	77.2	97.2	122.5	154.3	309	617	1235
Trans/Chip (millions)	97	122	153	193	243	307	386	773	1546	3092
Chip Size (Sq. Cm.)	140	140	140	140	140	140	140	140	140	140
High-Performance MPU										
Transistors/Chip (millions)	276	348	439	553	697	878	1106	2212	4424	8848
Chip Size (mm2)	310	310	310	310	310	310	310	310	310	310
Pins	1200	1320	1452	1600	1760	1936	2140	2782	3616	4702
Pkg Cost (cents/pin)	1.20	1.14	1.08	1.03	0.98	0.93	0.88	0.84	0.80	0.76
Ghz	1.68	2.32	3.09	3.99	5.17	5.63	6.74	11.51	19.35	28.75
Mask Levels	25	25	25	25	25	27	27	27	29	29

Exhibit 6: Table of Capital Cost of DRAM Fab and Feature Size

Year	DRAM Fab Cost $ Millions	DRAM Line Width (nm)
1970	6	
1975	20	
1977	30	
1980	50	
1982	100	
1983	160	
1985	200	
1986		1200
1987	333	
1988		800
1989		
1990	450	
1991	600	500
1992		
1993		
1994	800	
1995	900	
1996		300
1997	1200	230
1998	1500	
1999		180
2000	1650	150
2001	1900	130
2002	2500	110
2003		90

Source: IC Knowledge, VLSI, Intel

Exhibit 7a: DRAM Capital Cost Trend

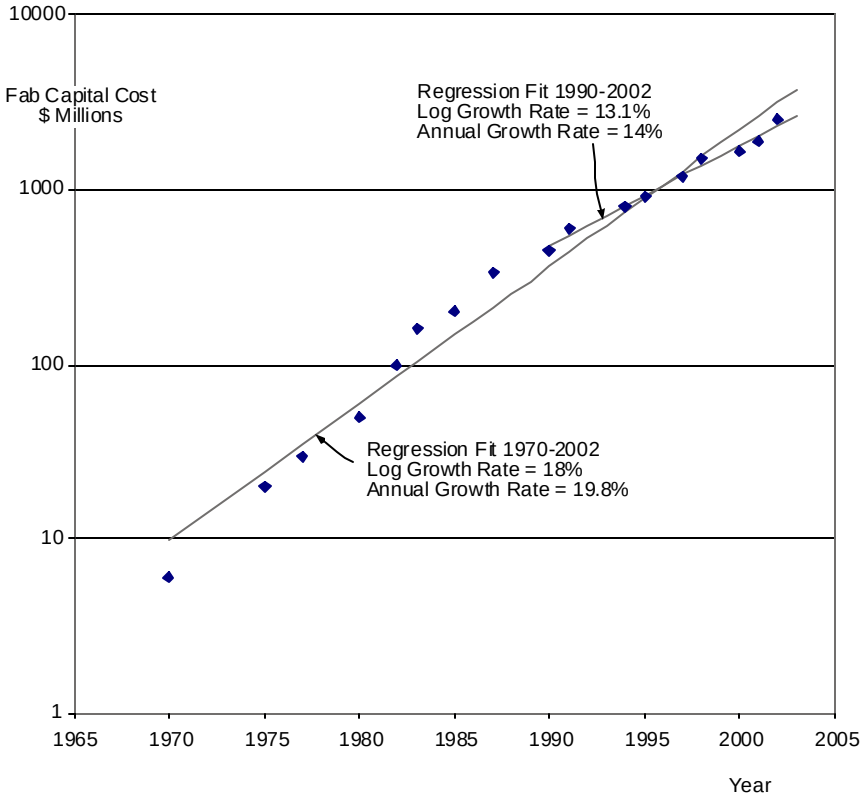


Exhibit 7b: DRAM Feature Size Trend

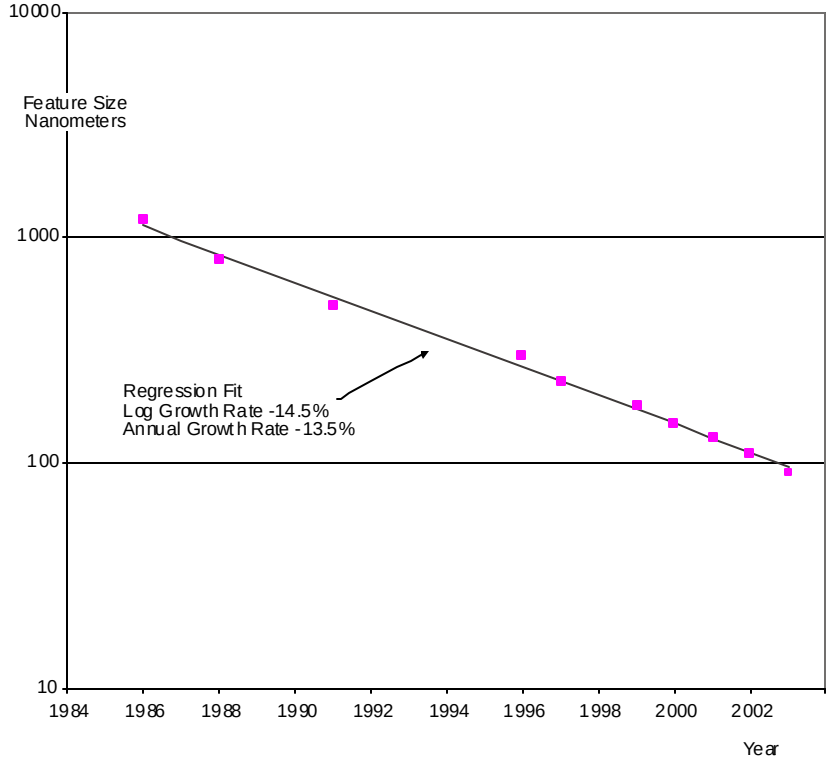
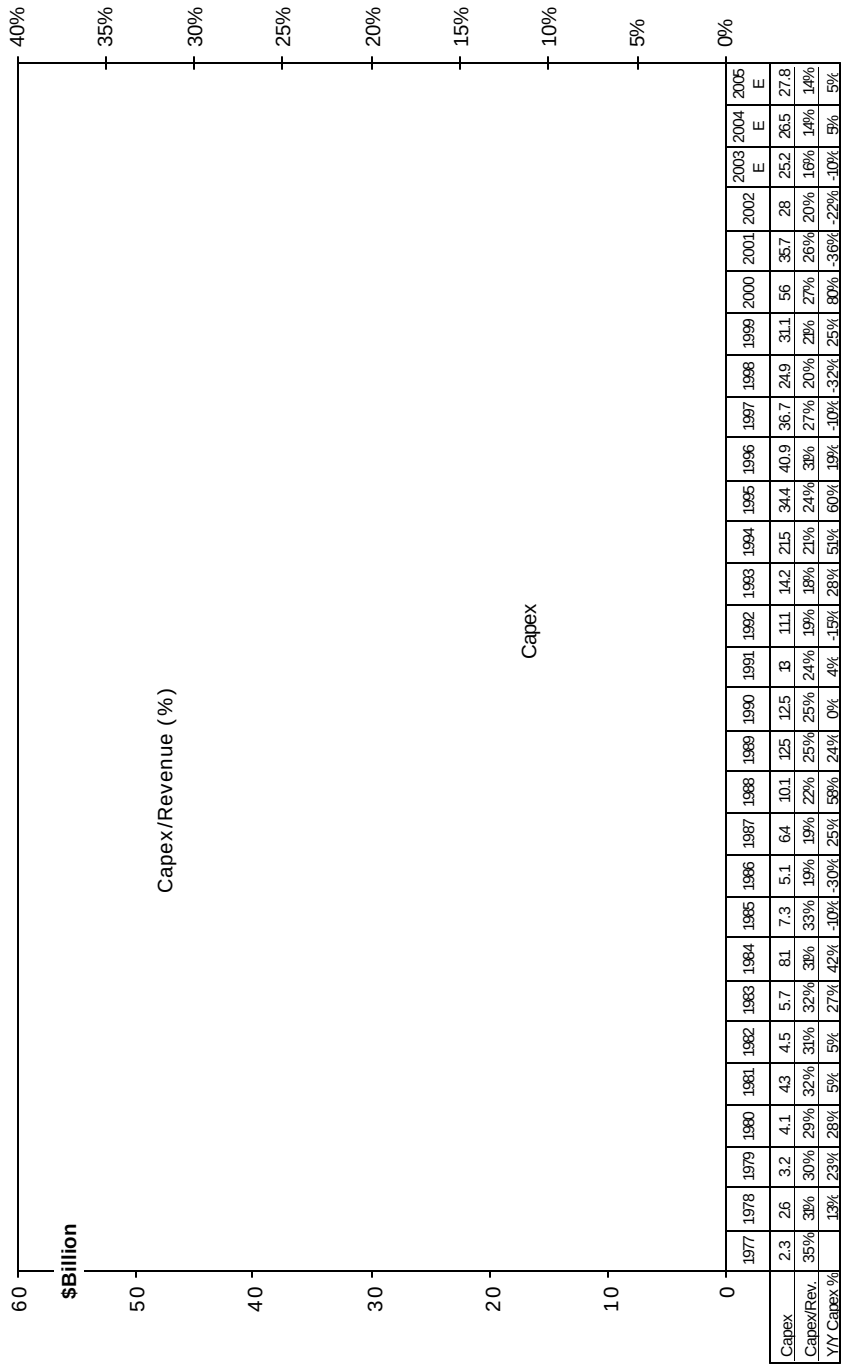


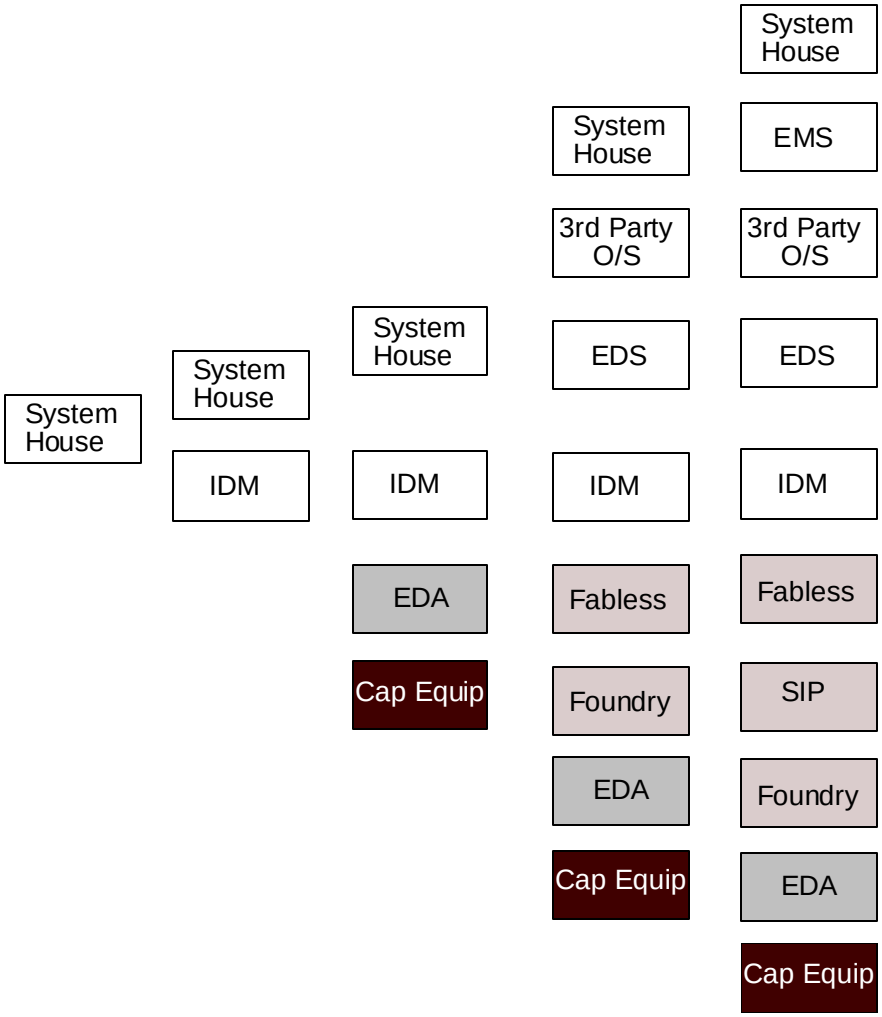
Exhibit 8: Capital Expenditures by IC Makers



	1977	1978	1979	1980	1981	1982	1983	1984	1985	1986	1987	1988	1989	1990	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003 E	2004 E	2005 E
Capex	2.3	2.6	3.2	4.1	4.3	4.5	5.7	8.1	7.3	5.1	6.4	10.1	12.5	12.5	13	11.1	14.2	21.5	34.4	40.9	36.7	24.9	31.1	56	35.7	28	25.2	26.5	27.8
Capex/Rev.	35%	31%	30%	29%	32%	31%	32%	31%	33%	19%	19%	22%	25%	25%	24%	19%	18%	21%	24%	31%	27%	20%	21%	27%	26%	20%	16%	14%	14%
Y/Y Capex %		13%	23%	28%	5%	5%	27%	42%	-10%	-30%	25%	59%	24%	0%	4%	-15%	28%	51%	60%	19%	-10%	-32%	25%	80%	-36%	-22%	-10%	9%	5%

Source: Adapted from JMP Securities, “Semiconductors: The Long and Winding Road to Recovery,” Nov. 5, 2002

Exhibit 9: Industry Disaggregation



Source: Merrill Lynch

KEY:

IDM	=	Integrated Device Manufacturer (Intel, STMicroelectronics, Micron)
EDA	=	Electronic Design Automation (Synopsis, Mentor, Cadence)
EDS	=	Electronic Design Services (Cadence-Tality, Old part of Parthus, S3 (Philips), New Logic)
3rd party O/S	=	Windriver, Symbian, PalmOS, Microsoft
SIP	=	Semiconductor intellectual property (ARM, MIPS, Tensilica)
Fabless	=	QUALCOMM, Xilinx, Altera, NVidia
Foundry	=	TSMC, UMC, Chartered